

Konrad Odell

Healthcare Data Platforms & Agentic AI Systems

I've been shipping software for twenty years. Since 2022 I've been on the architecture team at Moxe Health, a HIPAA-regulated clinical data exchange. The work is data platform and integration: Databricks lakehouse and Unity Catalog governance, FHIR pipelines, and for the last two years, evaluating LLM and agent approaches against real clinical and billing workloads: proofs of concept that had to prove their value before anyone built on them, and a clinical document generation pipeline.

On my own time I built and run ElectricityFinder, a paid subscription product for people shopping Texas retail electricity plans. It runs an LLM extraction pipeline in production behind an evaluation gate, plus the compliance work a regulated market requires. I also maintain the tooling I build with: MCP servers, a library of agent skills, and evaluation harnesses that decide what is allowed to ship.

The way I build with coding agents is a worker and verifier loop: one agent does the work, a separate one with fresh context has to prove it runs and does the thing before I accept it, and the ones that can't be verified don't ship. Most of my tooling exists to make that loop cheap, because a system that fails quietly costs more than one that breaks loudly.

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Work **Moxe Health**

● Staff Engineer, Architecture

Jan 2024 – Present

Architect for a B2B healthcare data interchange platform (HIPAA-regulated clinical data at scale), focused on data-platform capability, applied-AI evaluation, and cross-team technical coordination.

- Identified, scoped, and validated applied-AI opportunities against real clinical and financial workloads, including a multimodal LLM proof-of-concept that surfaced a material volume of recoverable invoice value
- Designed a FHIR-to-template clinical document generation pipeline
- Built a proof of concept that turns FHIR chart data into analyst-ready extracts, validated end-to-end against a simulated FHIR harness rather than by inspection
- Partnered on Databricks implementation including Unity Catalog governance, lakehouse architecture, and self-service tooling for data analysis
- Coordinated architecture reviews for a large-scale clinical lakehouse migration off a legacy external provider
- Documented module boundaries and state diagrams for monolith decomposition planning on a core clinical data processing system
- Coordinated architecture decisions across data platform, clinical retrieval, and revenue-operations functions, presenting tradeoffs to both engineers and senior leadership
- Coached senior engineers on architectural thinking and mentored across disciplines, including a formal data-science mentee

• Senior Software Engineer / Team Lead

Sep 2022 – Jan 2024

Led the consent-management service team on a HIPAA-regulated B2B clinical data interchange platform; owned delivery, deploy safety, and the service architecture.

- Led consent-management service team (Java, Spring Boot, NATS) with zero-downtime blue/green deployment
- Built deploy-time safety mechanisms: canary deployment, smoke tests, and auto-rollback
- Ran the team's delivery cadence and reviews; the operational patterns (canary, smoke tests, auto-rollback) became the platform-wide integration-testing standard

Senior Software Engineer II

Cognizant Softvision → Discover Financial Services

Aug 2021 – Sep 2022

Built card-tracking API ingesting USPS and internal events on Spring Boot microservices in a regulated financial-services environment.

- Introduced contract-test pipeline (WireMock + Pact) to catch breaking API changes pre-production
- Partnered with DevOps on Helm-based release process, reducing deploy time from 40 to 8 minutes
- Mentored junior engineers on testing practices and senior engineers on cross-team communication

Senior Software Engineer → VP of Research

Nearstar Inc.

Sep 2010 – Aug 2021

Legacy modernization, Docker adoption, CI/CD implementation, and performance optimization; grew from senior IC into a research/architecture leadership role.

- Dockerized monolith suite enabling per-customer configuration and a SaaS pivot
- Optimized multithreaded PDF loader during COVID spike: +260% throughput, backlog cleared in 36 hours
- Designed zero-downtime patch mechanism using Jenkins-orchestrated blue/green deployment
- Migrated Windows servers to Linux (MariaDB), enabling zero-downtime deployments and improved performance

Software Developer

ClubCorp

Jun 2001 – Jul 2008

Mainframe to Oracle migration; built authentication framework with salted password hashing.

Education

UT Dallas

B.S. in Computer Science

Projects

ElectricityFinder

- LLM plan-document extraction pipeline gated by an automated eval flywheel — extractions that fail the harness cannot reach production
- Full stack solo: Cloudflare Workers + D1, React/TypeScript frontend, Stripe subscriptions, Smart Meter Texas usage integration
- Registered PUCT electricity broker (BR260062); PUCT disclosure, CAN-SPAM, and TDPSA retention compliance designed in from the start
- ~5,000 automated tests, deploy-time bundle invariants, and post-deploy content smoke checks that assert rendered results, not just HTTP status

LLM Extraction

Eval-Gated Deploys

Cloudflare Workers

D1

TypeScript

Stripe

Regulated Domain

Product Ownership

Agentic Engineering Estate

- Reusable building blocks: a library of 68 agent skills (usage-instrumented and pruned by actual fire-count), 21 deterministic guard hooks, MCP servers and tool adapters in daily production use, and sub-agent/workflow orchestration patterns
- Custom eval harnesses rather than vibes: an own-commit SWE-bench-style benchmark for scoring local and frontier models on real repository work, plus adversarial-eval and domain-specific benches
- Deterministic orchestration with worker/verifier separation; autonomous overnight loops shipped 8 independently fix-verified production bug fixes

unattended

- Belief-state engine pairing LLM extraction with deterministic code judges, over a hybrid lexical + semantic search layer
- Multi-provider routing in practice (Anthropic, Google, and local open-weight models behind one gateway), plus local serving and LoRA fine-tuning on consumer hardware

Claude Code

Agent SDK

MCP

Multi-Agent Orchestration

Eval Harnesses

LLM-as-Judge

Python

llama.cpp

SchoolBrain

PostgreSQL

OAuth

Gemini

FastAPI

Next.js

RecipeBrain

Supabase

FastAPI

PostgreSQL Vector

Embeddings

Publications

I Built a Factory Control Plane for AI Agents in One Afternoon

Published by konradodell.com

Mar 2026

A human control surface over a seven-phase agent pipeline (planner, worker, reviewer, verifier, monitor, reporter), built in one Claude Code session; argues the deterministic layer is the control plane, not scaffolding to discard.

Stop Getting Attached to Your AI Tooling

Published by konradodell.com

Feb 2026

Coordinating six parallel Claude Code sessions with an orchestrator/worker file protocol, and keeping that tooling disposable as the models change.

The Rutter Pattern: AI Memory Through Continuous Refinement

Published by konradodell.com

Dec 2025

Two-tier agent memory: per-session logs distilled into durable patterns only once they prove themselves.

Skills

Applied AI & Agents

Production LLM pipelines

Multi-agent orchestration

Evaluation & Reliability

Custom eval harnesses

LLM-as-judge gates

AI coding agents as production build tools

MCP servers & tool integrations

Reusable agent skills / sub-agents

Retrieval & hybrid semantic search

Multi-provider model routing

Local inference & LoRA fine-tuning

Healthcare & Regulated Domains

HIPAA FHIR HL7

quality measures

Clinical data processing

PII governance

PUCT / CAN-SPAM / TDPSA

Data Platforms

Databricks (Unity Catalog, Genie, Lakehouse)

Trino Iceberg dbt DuckDB

Vector databases

DevOps

Docker Kubernetes

GitHub Actions

Blue/green & canary deploys

Adversarial evaluation

Worker/verifier separation

Deploy-time content smoke tests

Mutation & contract testing

Languages

Python TypeScript Java

SQL JavaScript Bash

Cloud & Distributed Systems

AWS (API Gateway, Lambda, EKS, RDS, S3)

Cloudflare (Workers, D1, Tunnels)

Event-driven architecture (NATS)

Enterprise integration

Data pipelines